SCAD-Net: A Semi-Supervised Connectivity-Aware Decoupling Network for High-Resolution Remote Sensing Image Change Detection

Yuling Zhou, Kun Tan, *Senior Member, IEEE*, Xue Wang, and Wen Zhang

***Abstract*—Semi-Supervised change detection (SSCD) in remote sensing faces two critical challenges: cyclical error propagation from noisy pseudolabels and the loss of fine-grained boundary details. To address these, we propose a novel SSCD framework, the semi-supervised connectivity-aware decoupling network (SCAD-Net). SCAD-Net breaks the cycle of error amplification with a strategy called pixelsregions via curriculum-guided consistency learning (P2R-CL). This strategy progresses from initial, high-confidence pixel-level supervision to more flexible, region-based semantic correction as training matures. To tackle boundary ambiguity, SCAD-Net employs a two-stage decoupling architecture: a channel information decoupling module (CIDM) separates categorical and directional features, followed by a multiscale attention and feature integration (MAFI) module that reintegrates this information for precise boundary localization. Experiments on four benchmark datasets demonstrate our method's superiority. On the Shanghai Dataset (SHD), using only 10% labeled data, SCAD-Net achieves an F1-score of 90.1%, representing a 3.88% improvement over state-of-the-art methods. Similarly, strong performance is observed on the LEVIR-CD (90.31% F1), WHU-CD (89.35% F1), and CDD (87.05% F1) datasets. This feature decoupling paradigm offers a robust approach for semi-supervised remote sensing in environments.**

***Index Terms*—Change detection (CD), edge information, remote sensing, semi-supervised learning (SSL).**

I. Introduction

CHANGE detection (CD) is a critical area of remote sensing applications, which is widely utilized in urban planning, ecological, and environmental protection [1], [2], agricultural assessment [3], and other domains. The primary goal of CD is to identify land-cover changes within the same area from bitemporal image pairs. With ongoing advancements in the spatial and temporal resolutions of remote sensing imagery, high-resolution images now offer more comprehensive and detailed insights into land-cover characteristics. This improvement significantly enhances the accuracy and timeliness of CD, providing robust support for its diverse applications.

The traditional CD methods typically rely on handcrafted feature extraction techniques to detect land-cover changes. These methods include algebra-based image differencing [4], image quantization [5], transformation-based principal component analysis (PCA) [6], and Kauth–Thomas transformation (KT) [7] methods, as well as classification-based support vector machine (SVM) [8] and random forest (RF) [9]. While these methods demonstrate effectiveness in certain specific scenarios, they often face challenges such as insufficient robustness and limited feature representation capabilities. In recent years, deep learning-based CD methods have emerged as a dominant research direction. Deep learning has been integrated into CD mainly as a feature extractor embedded in the traditional process of CD [10], [11], [12]. Compared with the traditional methods, these methods offer notable advantages, including a higher accuracy, enhanced noise reduction capabilities, and improved generalization performance, with the model structures generally categorized into four types: 1) convolutional neural network (CNN)-based structures; 2) transformer-based structures; 3) Mamba-based structures; and 4) hybrid structures. CNN-based CD models, such as UCDNet [13], Double U-Net [14], and MSP-CD [15], leverage fully convolutional networks to generate pixel-level predictions for high-resolution images, enabling accurate detection of binary and target changes by capturing both global and local features [16], [17], [18]. Transformer-based models, such as TCIANet [19], D-LIEG [20], and SCDVit [21], leverage the basic transformer module to extract semantic features, offering promising performances by modeling dense pixel relationships, although they are often computationally intensive. Recently, Mamba-based models, such as ChangeMamba [22], FEMCD [23], and CDMamba [24], have emerged as efficient alternatives that effectively capture long-range dependencies while maintaining computational efficiency. Furthermore, other variants such as CD-STMamba [25] and MSA [26] have demonstrated competitive performance, particularly in spatiotemporal feature extraction. Hybrid structures have been extensively explored, including models such as ChangeFormer [27], ConvTransNet [28], and SMNet [29],

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Yuling Zhou, Kun Tan, and Xue Wang are with the Key Laboratory of Geographic Information Science, Ministry of Education, East China Normal University, Shanghai 200241, China (e-mail: yeah.yuling@gmail.com; tankuncu@gmail.com; wx_ecnu@yeah.net).

Wen Zhang is with Shanghai Municipal Institute of Surveying and Mapping, Shanghai 200063, China (e-mail: 373743929@qq.com).

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along with more recent CNN–Mamba combinations such as CWMamba [30] and SPMNet [31]. This trend toward sophisticated feature extraction and fusion is not unique to CD; it mirrors broader advancements across remote sensing for various dense prediction tasks. For instance, recent works have explored spatiotemporal fusion for CD [32], dual-domain networks for hyperspectral image super-resolution [33], hyperspectral–multispectral image fusion [34], [35], and even hypergraph-based transformers for high-resolution semantic segmentation [36]. These advanced models underscore the rapid progress in the field but also highlight their shared reliance on high-quality supervision, which we will discuss next.

While supervised learning methods can deliver excellent performances in CD tasks, they are heavily reliant on high-quality labeled data, which limits their scalability. This challenge is not unique to remote sensing but a pervasive issue in computer vision, motivating the exploration of data-efficient learning paradigms [37], [38], [39], [40], [41]. In the context of remote sensing, this data dependency presents a formidable bottleneck. The process of labeling remote sensing imagery is labor-intensive, costly, and susceptible to the subjectivity and complexity inherent in annotation. Consequently, one promising direction to circumvent these challenges is semi-supervised learning (SSL), particularly through pseudolabeling. To overcome this challenge, researchers have increasingly adopted SSL approaches, which aim to maximize the utility of unlabeled data while depending on a limited set of labeled samples. This reduces the dependence on manual annotation and enhances both the learning efficiency and generalization capability of the models. The semi-supervised techniques can generally be categorized into three categories: pseudolabel-based methods [42], adversarial learning-based methods [43], and consistency learning-based methods [44]. Pseudolabeling involves using the model to assign labels to unlabeled data, which are then used for network training. Although this approach is intuitive, model performance becomes heavily contingent on the quality of the generated pseudolabels. To address this vulnerability, recent studies have focused on more sophisticated sample selection strategies. To mitigate this, recent studies have explored more sophisticated sample selection mechanisms. For example, S2Match [45] introduced a self-paced learning paradigm to dynamically manage the use of unlabeled data, thereby mitigating the risk of the model being trapped. Adversarial learning, inspired by generative adversarial networks (GANs), improves the model's representation of unlabeled data by introducing two networks—a generator and a discriminator—which interact adversarially. Despite its effectiveness, adversarial learning usually requires high training stability. Consistent learning, on the other hand, represents a more robust approach by emphasizing consistency in the predictions across augmented versions of the same data. This is typically achieved by generating diverse augmented representations of unlabeled data and ensuring consistent training, thereby enhancing the model's generalization capability.

Consistency regularization methods, particularly those employing a teacher–student architecture like UniMatch [46], have demonstrated considerable efficacy in SSL. However, their direct application to remote sensing CD faces two fundamental and interconnected challenges.

The first challenge is about a methodological contradiction that undermines the learning process. The technique relies on aggressive data augmentations to learn robust features and yet enforces this learning through a rigid, pixelwise consistency loss that is fundamentally misaligned with the geometric shifts induced by such transformations. This conflict is exacerbated by static pseudolabeling thresholds that discard ambiguous yet informative samples. Consequently, the model becomes prone to generating noisy and unreliable supervisory signals, ultimately destabilizing the training process. Moreover, the network architecture is tasked with a dual objective of semantic discrimination and geometric precision. It must simultaneously suppress spurious changes from irrelevant factors (e.g., illumination and seasonality) while delineating the exact boundaries of true changes. While this is a known challenge in fully supervised learning settings, its impact is critically amplified in a semi-supervised context. That is, any architectural inadequacy in boundary delineating generates structurally deficient pseudolabels, which are then fed back as training signals. This establishes a destabilizing feedback loop that systematically degrades the consistency learning process.

To address these parallel challenges, we have proposed the semi-supervised connectivity-aware decoupling network (SCAD-Net). SCAD-Net integrates novel connectivity-aware mechanisms designed specifically to enforce geometric precision and preserve boundary fidelity. By prioritizing the structural coherence of change regions, our approach not only achieves state-of-the-art performance with limited labeled data but also builds inherent robustness against the noisy signals generated by consistency regularization.

Therefore, the main contributions of this work are given as follows.

1) To resolve the architectural tension between semantics and precision, we propose a channel information decoupling module (CIDM). By explicitly modeling spatial connectivity, this module disentangles high-level semantic information from fine-grained boundary features, significantly enhancing the model's ability to achieve superior boundary fidelity in the final change map.
2) We propose a multiscale attention and feature integration (MAFI) module to intelligently reintegrate the semantic and boundary feature streams. By strategically fusing this information across multiple scales, the MAFI module constructs a comprehensive change representation that is both semantically rich and geometrically precise, improving detection accuracy for objects of varying scales.
3) To solve the methodological contradiction of consistency learning, we propose pixels–regions via curriculum-guided consistency learning (P2R-CL). This robust teacher–student strategy stabilizes training and builds resilience against noisy labels. It mitigates the impact of flawed pseudolabels by adaptively shifting the learning objective from unreliable individual pixels to the more stable semantic consensus of entire regions, guided by a progressive curriculum.

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4) We developed the Shanghai Dataset (SHD), which includes 25 600 pairs of high-resolution (0.1 m) images tailored for building CD. The superiority of the proposed method is demonstrated through extensive experimentation on four distinct datasets. The results confirm that our approach surpasses current state-of-the-art semi-supervised methods, showing marked improvements under various proportions of labeled data (5%, 10%, 20%, and 40%).

The structure of this article is outlined as follows. An overview of existing semi-supervised works and connectivity networks is provided in Section II. The proposed methodology, along with its foundational concepts, is described in Section III. A detailed analysis of the experimental results is the focus of Section IV. This article concludes with Section V, which summarizes our contributions and explores potential avenues for subsequent research.

II. Related Work

This section presents a detailed review of existing studies on semi-supervised CD (SSCD). The related works can be grouped into two primary categories: 1) semi-supervised methods for remote sensing CD and 2) connectivity networks.

A. Semi-Supervised Methods for Remote Sensing CD

Annotating remote sensing imagery is a challenging task, particularly for large-scale datasets. To tackle this problem, SSL has emerged as a key strategy, enhancing model performance by leveraging information from both labeled and unlabeled data sources. Among these techniques, adversarial learning has shown considerable promise in the context of semi-supervised CD. The approach increases the diversity of the training samples by generating pseudodata and enhances the CD performance through the discriminator's ability to differentiate real changes from generated ones. For instance, Gong et al. [47] proposed the generative discriminatory classification network (GDCN), which improves CD by combining labeled data, unlabeled data, and pseudodata generated by a generator to cotrain the discriminator, thereby generating a predictive change map.

Similarly, Peng et al. [48] designed SemiCDNet, which is a semi-supervised CD network based on a GAN architecture. The model ensures consistency between labeled and unlabeled data predictions while aligning feature distributions of entropy maps through two discriminators. Chen et al. [49] proposedinstance-level change augmentation (IAug), which enhances the model generalization by generating diachronic images containing diverse building changes through adversarial training and achieves natural fusion of buildings and backgrounds using context-aware blending. However, adversarial learning is highly dependent on the quality of the generator, and the performance gains are often constrained by the quality of the generated pseudosamples. In addition to adversarial learning, pseudolabel technology is another widely used semi-supervised technique. It expands the training dataset by generating high-confidence pseudolabels for unlabeled data, which are then combined with the labeled data for joint training. For example, reliable contrastive learning (RCL) [50] introduces a filtering mechanism to select only the most trustworthy pseudolabels for model training. Similarly, the SSCD method [51] adopts a similar principle, leveraging a weak-to-strong augmentation scheme where predictions from weakly augmented images are used to supervise the model's output on their strongly augmented counterparts. The semi-supervised adaptive ladder network (SSALN) [52] iteratively generates pseudolabels using implicit information, optimizing their reliability and accuracy with each iteration. Yang et al. [53] introduced cross pseudosupervision based on ensemble learning (ECPS), a novel semi-supervised strategy that merges the outputs from multiple CD models using integrated learning, thereby significantly improving the pseudolabel quality and overall model performance.

Since pseudolabeling is essentially an SSL method based on hard labels, its core lies in assigning deterministic labels to unlabeled data. In contrast, consistency learning adopts a more flexible and dynamic learning strategy, focusing on optimizing the model's output consistency under different data augmentation conditions, rather than relying on fixed labels. This approach offers a more robust and adaptive solution to SSL. The classical FixMatch method [54] employs weakly augmented results to supervise strongly augmented predictions, providing a simple and efficient implementation of consistency learning. Shu et al. [55] developed a multitask CD framework comprising three branches: T1, T2, and the change branch. The Diff-T1 branch is constructed by calculating the difference between the change branch and the T2 branch. By enforcing the prediction consistency between the original T1 branch and the Diff-T1 branch, the network's generalization ability is effectively enhanced. Han et al. [56] proposed C2F-SemiCD, which employs consistency regularization to minimize the difference between the predictions of a student model and its corresponding teacher model, thereby improving the feature learning capacity for both labeled and unlabeled data.

Although the existing SSL methods have achieved remarkable results, they still face challenges such as limited feature expressiveness and poor model generalization in the case of extremely scarce labeled data. In the consistency regularization-based approaches, designing effective data perturbation strategies to enhance the model robustness and generalization remains a key issue. Therefore, exploring more effective consistency regularization strategies to further enhance model performance holds significant research value.

B. Connectivity Networks

Connectivity networks represent a class of methods designed to enhance feature representation by explicitly modeling the connectivity relationships between pixels or regions. The core concept involves constructing an eight-channel connectivity mask for the central pixel within an eight-neighborhood system, which captures connected features in various directions and facilitates the interaction of directional information. As a result, connectivity networks demonstrate strong modeling capabilities in visual tasks such as semantic

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segmentation and instance segmentation. Kampffmeyer et al. [57] proposed a pure connectivity network (ConnNet), which improves the overall semantic coherence and region smoothness by predicting the probability of each pixel's connectivity to its neighbors, thereby leveraging the multiple cascading contexts and remote pixel relationships embedded in the imagery. Yang and Farsiu [58] proposed segmentation based on a directional connectivity network (DconnNet), which separates the orientation subspace from the shared potential space, thereby enhancing the accuracy and orientation consistency of medical image segmentation.

In the domain of remote sensing analysis, connectivity principles are particularly relevant due to the inherent characteristics of remote sensing imagery. On the one hand, many critical land-cover features—such as roads, rivers, and building outlines—are inherently linear or areal continua whose topological integrity is essential for accurate segmentation. On the other hand, standard convolutional networks often produce fragmented segmentation maps when confronted with common challenges in high-resolution imagery, including occlusions (e.g., from trees or shadows) and spectral confusion. Connectivity networks address these challenges by introducing structural priors that implicitly or explicitly model spatial topology to generate topologically coherent results.

This approach has been successfully applied to various static scene segmentation tasks in remote sensing. In linear feature extraction, researchers have focused on resolving connectivity disruptions caused by occlusions. For instance, Xu et al. [59] and Mei et al. [60] effectively preserved target connectivity by capturing long-range contextual information from multiple directions and strengthening relationships between adjacent pixels. Similarly, Gao et al. [61] designed a network based on connectivity arrays and direction graphs to exploit the linear characteristics of road targets fully. For fine-grained segmentation of areal features such as buildings, Li et al. [62] and Zhou et al. [63] reframed the segmentation task as a connectivity prediction problem, incorporating direction-aware attention modules to reinforce the topological consistency of boundaries. Furthermore, Yang et al. [64] introduced a bilateral voting module to refine connectivity maps and improve the supervision mechanism for pixel linkages. These studies collectively demonstrate that explicitly modeling connectivity significantly enhances topological quality and boundary accuracy for both open linear structures and closed areal objects.

Despite its proven efficacy in remote sensing segmentation, the potential of connectivity modeling in CD remains largely underexplored. CD tasks face analogous challenges, ensuring that detected change regions are internally complete, exhibit well-defined boundaries, and maintain topological correctness. This is particularly critical when addressing subtle, fragmented, or geometrically regular changes, such as new road construction or building expansions, where capturing precise boundaries and topological structure is paramount. To address this issue, we propose the CIDM module, which models pixel neighborhood relationships to decouple categorical and directional features, thereby ensuring the boundary continuity and structural integrity of changed regions.

III. Method

In this section, we introduce the proposed SCAD-Net framework. The overall framework of the proposed method is illustrated in Fig. 1. SCAD-Net utilizes DeepLabv3+ as its backbone to extract multiscale contextual features from high-resolution remote sensing images. Building on this foundation, the proposed CIDM and MAFI modules are integrated to enhance the edge representation and suppress pseudochanges, thereby addressing the challenges of semi-supervised CD.

A. Channel Information Decoupling Module (CIDM)

To enhance the extraction of fine-grained edge information, we propose the CIDM (see Fig. 2). In the building CD task, since building contour lines are usually straight, the vertical and horizontal neighborhood information is sufficient to capture most of the edge information. Based on the aforementioned analysis, we propose a method to simplify the eight-neighborhood system of the connectivity network to a four-neighborhood system. This approach directly captures directional connectivity features through the network structure, eliminating the need for predefined connectivity masks, thereby reducing the computational complexity and effectively separating category-related features from directional features.

A channel mapping operation is first applied to reduce the input feature map to 4 × nclass (nclass = 2) using 3 × 3 convolution, and then, interpolation is performed to match the original resolution. Subsequently, the reduced feature maps are recoded by 1 × 1 convolution to further extract directional information. Connectivity vectors are then generated using global average pooling of the aggregated directional features to encapsulate the feature map's key directional information

$$F_{\text{mapped}} = \text{ReLU}\left(\text{BN}\left(W_1 \times F + b_1\right)\right) \tag{1}$$

$$F_{\text{unsampled}} = \text{Interpolate}\left(F_{\text{mapped}}, \text{scale_factor} = 32\right) \tag{2}$$

$$F_{\text{reencoded}} = W_2 \times F_{\text{unsampled}} + b_2 \tag{3}$$

$$d_{\text{prior}} = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} F_{\text{reencoded}}\left(i, j\right) \tag{4}$$

where the feature map $F \in \mathbb{R}^{C \times H \times W}$, and H' and W' denote the height and width of the recorded feature map, respectively. W_1, W_2, b_1, and b_2 are learnable parameters; BN denotes the batch normalization operation; and a rectified linear unit (ReLU) is the activation function. The module's design follows a clear logic: an information bottleneck (1) compels the network to learn a compact set of specialized directional feature detectors. These learned concepts are then projected back to the full resolution (2) and refined pixelwise (3). Finally, global average pooling (4) abstracts the spatial features into a global directional vector, d_{prior}. This vector provides a powerful statistical prior on the scene's dominant orientations, enabling subsequent modules to better distinguish the regular, geometric edges of buildings from irregular, noisy textures, thereby significantly improving the precision of CD. This

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method can effectively extract and integrate the directional information to provide richer feature representations for the subsequent tasks.

To effectively disentangle change information and directional information in the feature maps, we stratify both the connectivity vectors and feature maps into four slices based on the channel hierarchy. The connectivity vectors, as low-level linear combinations of highly discriminative directional features, ensure that each channel slice captures unique directional characteristics. Conversely, the feature map primarily contains fewer discriminative change features, enabling the feature slices to preserve essential information without compromising classification performance. By fusing the change and directional features from different channel slices and emphasizing key directional information through an adaptive weight coupling mechanism, this approach significantly enhances the edge extraction capabilities. The feature slices can be obtained using the following equation:

$$\text{feat}_i = \alpha_i \times \varphi\left(\text{DFRE}\left(\text{slice}_d, \text{slice}_c\right)\right)(i = 1, 2, 3, 4) \quad (5)$$

where $\text{slice}_d \in \mathbb{R}^{C/4\times1\times1}$ and $\text{slice}_c \in \mathbb{R}^{C/4\times H\times W}$ denote channel slices and feature slices, respectively, and $\alpha \in (0, 1)$ denotes the adaptive weights that balance the original and fused skip connections. φ is applied with a dropout probability of 0.5. Directional fusion re-encoding (DFRE) re-encodes these channel and feature slices, selectively emphasizing or suppressing direction-specific change features. The perceptual field of the feature slice_c is expanded using dilated convolution, after which the feature slice is processed through a channel prior convolutional attention (CPCA) [65] to generate the enhanced feature $R'_c \in \mathbb{R}^{C/4\times H\times W}$. This operation enhances the contextual correlation while retaining the original features. Summing R'_c with the feature slice_c produces enriched feature information $R_c \in \mathbb{R}^{C/4\times H\times W}$, which preserves the original features and adapts to edge enhancements. Subsequently, an activation function is applied to the channel slice_d, and the result is multiplied by R_c, selectively emphasizing or suppressing key informative features in specific directions. Finally, a dropout operation is applied to enhance the generalization capability, yielding edge-enhanced features $R \in \mathbb{R}^{C/4\times H\times W}$, as follows:

$$R_c = R'_c + \text{slice}_c = \text{CPCA}\left(\text{Conv}_{\text{dila}}\left(\text{slice}_c\right)\right) + \text{slice}_c \quad (6)$$

$$R = \varphi\left(\text{RELU}\left(\text{slice}_d\right) \times R_c\right) \quad (7)$$

where $\text{Conv}_{\text{dila}}()$ is a dilated convolution with a dilation rate of 2 and φ is a dropout operation with an application probability of 0.1.

B. Multiscale Attention and Feature Integration (MAFI) Module

Effectively fusing semantic and edge information while capturing both local and global information dependencies is a persistent challenge in computer vision. The traditional methods often struggle to fully leverage global contextual information while maintaining fine-grained local features. Simple feature concatenation or summation is insufficient for seamlessly integrating semantic and edge information, necessitating a more advanced and intelligent fusion strategy. To address this challenge, we propose the MAFI module (see Fig. 3).

The input to the MAFI module incorporates two types of features—semantic features $M_S \in \mathbb{R}^{C\times H\times W}$ and edge features $M_e \in \mathbb{R}^{C\times H\times W}$ by pixel-level addition to obtain $M \in \mathbb{R}^{C\times H\times W}$, incorporating semantic and edge information. To capture fine-grained local dependencies, a local attention block is introduced, consisting of two 1×1 convolutional layers containing an ReLU activation function and batch normalization, and feature extraction is performed using M to obtain $m_L \in \mathbb{R}^{C\times H\times W}$. To capture the long-range global dependencies, a global attention block is constructed. This block applies global mean pooling, followed by 1×1 convolution and an ReLU activation function, to obtain $m_G \in \mathbb{R}^{C\times H\times W}$, which is the aggregated global contextual information.

To fuse multiscale semantic and contextual information, four context blocks are proposed. These context blocks apply average pooling (for semantic contexts) and maximum pooling (for texture contexts) to downsample the input features M to 4×4, 8×8, 16×16, and 32×32 resolutions. The output of each context block is denoted as s_4, s_8, s_{16}, and s_{32}, and the corresponding texture pooling features are denoted as e_4, e_8, e_{16}, and e_{32}, where each s_i, e_i corresponds to a specific resolution. The integrated features M_F can be obtained using the following equations:

$$s'_{16} = \text{Gonv}\left(s_{16} + e_{16} + \text{NI}_{16}\left(s_{32} + e_{32}\right)\right) \quad (8)$$

$$s'_8 = \text{Gonv}\left(s_8 + e_8 + \text{NI}_8\left(s'_{16} + e_{16}\right)\right) \quad (9)$$

$$s'_4 = \text{Gonv}\left(s_4 + e_4 + \text{NI}_4\left(s'_8 + e_8\right)\right) \quad (10)$$

$$\varepsilon = \text{Sigmoid}\left(S_F\right) = \text{Sigmoid}\left(\text{NI}_{\text{HW}}\left(s'_4, s'_8, s'_{16}\right) + m_L + m_G\right) \quad (11)$$

$$M_F = M_s \times \varepsilon + M_e \times (1 - \varepsilon) \quad (12)$$

where Gonv() is a gated nonlocal block containing two 1×1 convolutional layers, with an ReLU() activation function and batch normalization in the middle. $\text{NI}_4(), \text{NI}_8()$, and $\text{NI}_{16}()$ represent the nearest neighbor interpolation, and the subscripts represent the scale factors of the interpolation. The final output M_F can effectively integrate local and global contextual information, enhancing the network's ability to capture fine-grained details and long-range dependencies.

C. P2R-CL: Pixels–Regions via Curriculum-Guided Consistency Learning

To address the training instability caused by unreliable pixel-level signals in standard consistency regularization, we propose a novel framework named P2R-CL. The core philosophy of P2R-CL is twofold: 1) incorporating unlabeled data by dynamically refining pseudolabel quality and 2) enforcing the learning of robust, change-sensitive representations through a challenging multilevel perturbation-driven consistency task.

1) Foundational Stability via EMA Teacher: A primary source of instability in teacher–student frameworks is the potential for rapid, noisy updates to the teacher model. Rather than employing conventional fixed-weight teachers, P2R-CL employs a theoretically motivated exponential moving average (EMA) teacher, which is specifically designed for the high-noise, high-ambiguity characteristics of remote sensing data.

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TABLE I
PERTURBATION OPERATION SET

weak perturbation	Random scale, Random flip, Random crop
strong perturbation	Identity, Invert, Auto-contrast, Equalize, Gaussian blur, Contrast, Sharpness, Color, Brightness, Hue, Posterize, Solarize, CutMix

Our key insight is that this temporal smoothing provides a stable source of supervision, crucial for generating reliable pseudolabels for both broad regions and subtle pixel-level changes. To ensure initial stability, the entire framework is anchored by a supervised warm-up period of 15 epochs, ensuring the student model learns meaningful representations before engaging with pseudolabels. Subsequently, the teacher weights θ'_t are updated after each step

$$\theta'_t = \alpha_t \cdot \theta'_{t-1} + (1 - \alpha_t) \cdot \theta_t \tag{13}$$

where α_t is a cosine-annealed decay factor that adapts from faster updates early in training ($\alpha_{\text{start}} = 0.95$) to greater stability later ($\alpha_{\text{end}} = 0.999$)

$$\alpha_t = \alpha_{\text{end}} - (\alpha_{\text{end}} - \alpha_{\text{start}}) \times \frac{\cos(\pi t/T) + 1}{2}. \tag{14}$$

All pseudolabels for unlabeled data x_{u} are generated exclusively by the stable teacher model, which is crucial for providing reliable supervision for both large, homogeneous regions and ambiguous pixel-level boundaries

$$p_{\text{u}} = \arg\max\left(f_{\theta'}(x_{\text{u}})\right) \tag{15}$$

where p_{u} is the generated pseudolabel for the unlabeled input x_{u} and $f_{\theta'}$ denotes the teacher model with its EMA-updated weights θ'.

2) Adaptive Pseudolabel Gating for Progressive Learning: While the EMA teacher provides a stable source of supervision, the pseudolabels still vary in quality. To prevent the model from being corrupted by low-quality labels, we present a curriculum that strategically shifts the model's focus from easy regions to difficult pixels. To prevent model collapse from noisy supervision, we introduce an adaptive pseudolabel gating mechanism to tackle the significant risk in remote sensing due to registration errors, shadows, or seasonal variations that can mimic change. This employs a dynamic confidence threshold τ_t that handles the pseudolabel selection process

$$\tau_t = \tau_{\text{start}} - (\tau_{\text{start}} - \tau_{\text{end}}) \cdot \frac{t^\gamma}{T^\gamma}. \tag{16}$$

This mechanism employs a dynamic confidence threshold τ_t that polynomially decays (from $\tau_{\text{start}} = 0.98$ to $\tau_{\text{end}} = 0.95$) to implement a region-to-pixel curriculum. Initially, the high threshold selects only "easy," high-confidence pseudolabels, which typically represent large, unambiguous regions. As τ_t lowers, the model is gradually exposed to "harder" samples corresponding to intricate boundaries and fine-grained pixel details, effectively shifting the learning focus from coarse regions to precise pixels.

3) Multilevel Perturbation for Robust Representation: With a mechanism in place to secure high-quality pseudolabels, the challenge shifts from what to learn to how to learn. To prevent the student from passively mimicking the teacher's outputs, P2R-CL introduces a multilevel perturbation strategy. It is specifically engineered to compel the student to transition from processing isolated pixels to understanding cohesive image regions. It forces the model to learn the invariant semantics of "change" by discovering that a pixel's identity is defined by its regional context, not by superficial visual cues.

At the input level, we enforce multiview consistency to build regional context awareness. For each unlabeled image x_u, a stable pseudolabel p_{u} is first generated by feeding a weakly augmented view $x_w = \mathcal{A}_w(x_u)$ to the teacher model. The student model is then compelled to reproduce consistent predictions on two independently and strongly augmented views, $x_{s1} = \mathcal{A}_s(x_w)$ and $x_{s2} = \mathcal{A}_s(x_w)$. The proposed strong augmentation $\mathcal{A}_s$ involves a stochastic composition of transformations (e.g., color jitter, Gaussian blur, and cutout; see Table I), which is critical for remote sensing data. By forcing consistency across these strategically perturbed views, we train the model to be invariant to nuisance variables, ensuring its decisions are grounded in the stable semantics of regions, not the volatility of individual pixels.

To complement the feature-level perturbations, we introduce a learnable feature-space distortion module (φ) within the student's forward pass. This module reinforces regional understanding at a deeper, more abstract feature level. Specifically, φ is implemented as a lightweight block comprising a dropout layer (probability 0.5) followed by a nonlinear transformation using two sequential 3×3 depthwise separable convolutions with GeLU activations. The forward pass for the student model on a strongly augmented input is, therefore,

$$F_{s1} = f(\theta)(\varphi(\delta(x_{s1}))) \tag{17}$$

$$F_{s2} = f(\theta)(\varphi(\delta(x_{s2}))) \tag{18}$$

where φ is the feature layer perturbation applying a dropout operation with a probability of 0.5 and δ is a nonlinear perturbation based on convolution operations. The synergistic interaction of perturbations at both the image and feature levels creates a highly challenging learning objective. It compels the model to distinguish between aleatoric uncertainty (inherent data variations) and epistemic uncertainty (the model's ignorance), ultimately fostering a more abstract and robust understanding of what constitutes a "change" in complex remote sensing scenes.

D. Loss Function

In order to train the model efficiently, an adaptive loss function was designed. The overall loss$L$ in the consistency regularization framework consists of two cross-entropy loss terms, including the supervised loss L_l and the unsupervised loss L_u. L_l is the cross-entropy loss between the model predictions of the labeled data and the mask. The total loss is defined as

$$L = \mu L_l + (1 - \mu) L_u \tag{19}$$

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TABLE II
QUANTITATIVE PERFORMANCE ON THE CDD DATASET

	5%			10%			20%			40%		
	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
AdvNet [66]	56.66	51.72	39.52	61.71	57.11	44.62	67.47	63.49	50.91	75.61	72.70	60.78
S4GAN [67]	67.15	63.60	50.54	71.82	68.66	56.04	76.75	74.05	62.27	81.38	79.17	68.61
RCL [50]	74.94	71.60	59.93	85.96	83.94	75.38	88.40	86.69	79.21	89.67	88.14	81.27
SemiCD [68]	81.33	78.22	68.54	85.43	83.45	74.57	89.03	87.41	80.23	90.27	88.86	82.27
FPA-SSCD [51]	81.36	78.93	68.58	85.81	83.90	75.15	88.75	87.18	79.78	90.09	88.67	81.97
C2F-SemiCD[56]	75.94	73.15	61.21	80.86	78.52	67.87	86.64	84.96	76.43	92.44	91.43	85.94
CutMix-CD[69]	80.41	77.72	67.23	85.16	83.06	74.15	89.76	88.29	81.42	92.10	90.96	85.37
SCAD-Net	**81.60**	**79.50**	**68.92**	**87.05**	**85.47**	**77.07**	**90.75**	**89.59**	**83.07**	**92.63**	**91.68**	**86.27**

TABLE III
QUANTITATIVE PERFORMANCE ON THE WHU-CD DATASET

	5%			10%			20%			40%		
	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
AdvNet [66]	51.72	50.10	39.05	65.63	62.07	43.35	74.84	73.67	49.30	77.44	76.41	60.48
S4GAN [67]	67.22	65.82	50.63	74.01	72.87	58.75	81.06	80.18	68.15	83.28	82.51	71.34
RCL [50]	67.82	66.56	51.35	83.62	82.86	71.84	84.70	82.87	73.45	89.40	88.86	80.83
SemiCD [68]	77.41	76.40	63.15	82.10	81.21	69.64	82.08	81.09	69.60	88.91	88.34	80.03
FPA-SSCD [51]	80.91	79.96	67.95	81.94	80.99	69.40	83.10	82.16	71.08	88.02	87.38	78.60
C2F-SemiCD[56]	84.55	83.95	73.24	88. 79	88.36	79.84	90.16	89.77	82.08	91.11	90.76	83.67
CutMix-CD[69]	87.56	87.08	77.87	88.71	88.26	79.71	88.83	88.40	79.90	92.01	91.70	85.19
SCAD-Net	**88.40**	**87.95**	**79.22**	**89.35**	**88.93**	**80.75**	**90.79**	**90.41**	**83.13**	**92.25**	**91.94**	**85.61**

TABLE IV
QUANTITATIVE PERFORMANCE ON THE LEVIR-CD DATASET

	5%			10%			20%			40%		
	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
AdvNet [66]	37.49	35.05	23.07	53.06	51.13	36.11	63.31	61.55	46.32	69.38	67.99	53.11
S4GAN [67]	36.10	34.59	22.03	64.59	63.13	47.70	73.74	72.50	58.40	78.72	77.62	64.91
RCL [50]	83.47	82.60	71.63	85.15	84.37	74.14	85.12	84.33	74.09	85.60	84.83	74.83
SemiCD [68]	82.09	81.39	69.62	86.73	86.07	76.57	86.96	86.30	76.93	87.69	87.06	78.08
FPA-SSCD [51]	85.31	84.58	74.38	87.05	86.39	77.07	87.71	87.08	78.11	87.93	87.31	78.46
C2F-SemiCD[56]	88.44	87.84	79.28	89.67	89.14	81.28	90.72	90.24	83.02	**90.96**	**90.48**	**83.42**
CutMix-CD[69]	87.74	87.08	78.15	88.91	88.32	80.03	89.53	88.98	81.05	89.70	89.15	81.32
SCAD-Net	**89.33**	**88.78**	**80.72**	**90.31**	**89.81**	**82.34**	**90.76**	**90.27**	**83.08**	90.43	89.94	82.53

where μ is the weight factor, which was set to 0.5 in this study. L_u aims to constrain the consistency between strong perturbation labels and weak perturbation labels. Consistency regularization relies on the idea that varying levels of perturbations applied to input samples should not cause significant changes in the model's output. Therefore, for the same objective, the results obtained by the model before and after applying perturbations should be consistent. The losses for the different perturbation flows can be calculated as follows:

$$L_w = L_{\mathrm{CE}}\left(p_w, \mathrm{mask}\right) + L_{\mathrm{CE}}\left(f_w, p_w\right) \tag{20}$$

$$L_{s1} = \frac{1}{B}\sum \left(\max\left(C_w\right) \geq \xi\right) L_{\mathrm{CE}}\left(p_{s1}, p_w\right) \tag{21}$$

$$L_{s2} = \frac{1}{B}\sum \left(\max\left(C_w\right) \geq \xi\right) \left(L_{\mathrm{CE}}\left(p_{s2}, p_w\right) + L_{\mathrm{CE}}\left(f_{s2}, p_w\right)\right) \tag{22}$$

where B represents the batch size for unlabeled data; ξ is a predefined confidence threshold used to filter out low-confidence, and thus potentially noisy, pseudolabels; C_w is an indicator function that activates the loss only if the model's highest class probability for a sample $\max(C_w)$ exceeds this threshold; p_w, p_{s1}, and p_{s2} represent the predictions of the perturbed data; and f_w and f_{s2} represent predictions obtained after applying an additional feature-level perturbation within the model. This is distinct from data augmentation and is achieved by applying techniques such as stochastic dropout to the intermediate feature maps before the final prediction layer. This encourages the model's feature representations to be robust, and L_{CE} denotes the cross-entropy loss. The

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TABLE V
QUANTITATIVE PERFORMANCE ON THE SHD DATASET

	5%			10%			20%			40%		
	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
AdvNet [66]	66.74	63.88	50.08	71.13	68.83	55.20	81.01	79.53	68.09	83.07	81.60	71.04
S4GAN [67]	70.88	68.91	54.89	70.92	69.00	54.95	78.79	77.26	65.00	82.33	80.91	69.97
RCL [50]	81.59	80.03	68.90	83.54	82.15	71.73	88.09	87.07	78.72	92.34	91.69	85.77
SemiCD [68]	75.25	73.35	60.33	83.85	82.55	72.19	87.22	86.16	77.34	91.56	90.86	84.43
FPA-SSCD [51]	76.91	75.22	62.48	82.64	81.29	70.42	87.01	85.99	77.01	92.48	91.83	86.01
C2F-SemiCD[56]	77.16	75.36	62.82	80.79	79.13	67.77	83.80	82.48	72.11	89.49	88.62	80.98
CutMix-CD[69]	85.43	84.29	74.57	86.22	85.11	75.78	90.83	90.06	83.20	90.60	89.81	82.81
SCAD-Net	**87.32**	**86.26**	**77.49**	**90.10**	**89.29**	**81.98**	**92.42**	**91.78**	**85.90**	**95.18**	**94.77**	**90.79**

unsupervised loss L_u is formulated as

$$L_u = \alpha L_w + \beta L_{s1} + \gamma L_{s2} \tag{23}$$

where α, β, and γ are the factors of the three parts of the formula, respectively, which are usually set to $\alpha = 0.5$, $\beta = 0.25$, and $\gamma = 0.25$. This choice places a higher emphasis on the stability of the weak-augmentation branch L_w, which is critical as it generates the pseudolabels for the other terms, while equally balancing the two strong-consistency losses that enforce robustness to more significant perturbations.

IV. EXPERIMENTS

This section describes how we evaluated the performance of the proposed method through extensive experiments. It covers the experimental setup, datasets, evaluation metrics, and ablation experiments.

A. Experimental Setup

1) Datasets: The WHU-CD dataset [70] from Wuhan University is a very-high-resolution (VHR) resource for CD. It is composed of two images of New Zealand from 2012 to 2016, each with a 0.2-m spatial resolution and a size of 32 507 × 15 345 pixels. In this study, all the images were cropped into nonoverlapping 256 × 256 pixel patches. The dataset was partitioned into training (5947 patches), validation (743 patches), and test (744 patches) sets.

The CDD dataset [71] offers a collection of seasonal CD images from Google Earth, notable for its inclusion of varied object sizes and seasonal shifts. It consists of eleven image pairs in total, with resolutions between 0.03 and 0.1 m. The original images came in two sizes: seven pairs at 4275 × 2700 and four at 1900 × 1000 pixels. The final processed collection contained 16 000 pairs of 256 × 256 images, partitioned as follows: 10 000 for training, 3000 for validation, and 3000 for testing.

The LEVIR-CD [72] dataset is a comprehensive resource for CD research. The dataset is focused on building growth and building decline. All the images were collected over an interval of between 5 and 14 years from Google Earth. The dataset consists of 637 image patches, each of size 1024 × 1024 pixels and a 0.5-m spatial resolution. Each image patch was segmented into small nonoverlapping patches of 256 × 256 pixels, which were divided into 7120 pairs for the training set, 1024 pairs for the validation set, and 2048 pairs for the test set.

The SHD dataset consists of 0.1-m resolution aerial images of Shanghai's Pudong New Area captured in 2016 and 2022 (see Fig. 5), covering an area of 16.78 km^2. This dataset primarily includes changes in buildings, such as additions, demolitions, and transformations. The original images, measuring 51 200 × 32 768 pixels, were partitioned into nonoverlapping patches of 256 × 256 pixels, resulting in 25 600 patches. The dataset was subsequently divided into training, validation, and test subsets with an 8:1:1 ratio, yielding 20 480, 2560, and 2560 patches, respectively.

2) Implementation Details: All the experiments were conducted using the PyTorch deep learning framework on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU with 24-GB CUDA memory. During training, each minibatch was composed of eight labeled images and eight unlabeled images, with the total number of epochs set to 100. The model was trained using the AdamW optimizer with an initial learning rate of 0.02 and a weight decay coefficient of 1e-4. The minimum learning rate was set to 10^{-6}.

3) Evaluation Metrics: To verify the performance of the proposed method, the precision (Pre.), recall (Rec.), F1-score (F1), overall accuracy (OA), kappa coefficient (KC), and Intersection over Union (IoU) are used here for the quantitative evaluation. Their definitions are given as follows:

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{24}$$

$$\text{Rec} = \text{TP}/(\text{TP} + \text{FN}) \tag{25}$$

$$F_1 = (2 \times \text{Pre} \times \text{Rec})/(\text{Pre} + \text{Rec}) \tag{26}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FN} + \text{FP}) \tag{27}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FN} + \text{FP}) \tag{28}$$

where TP, FP, FN, and TN represent true positive, false positive, false negative, and true negative, respectively.

B. Comparison With the State-of-the-Art Methods

To demonstrate the superiority of SCAD-Net, we compared it with five advanced semi-supervised CD methods: AdvNet [66], S4GAN [67], RCL [50], SemiCD [68], FPA-SSCD [51], C2F-SemiCD [56], and CutMix-CD [69], where AdvNet and S4GAN are based on adversarial learning, RCL is based on

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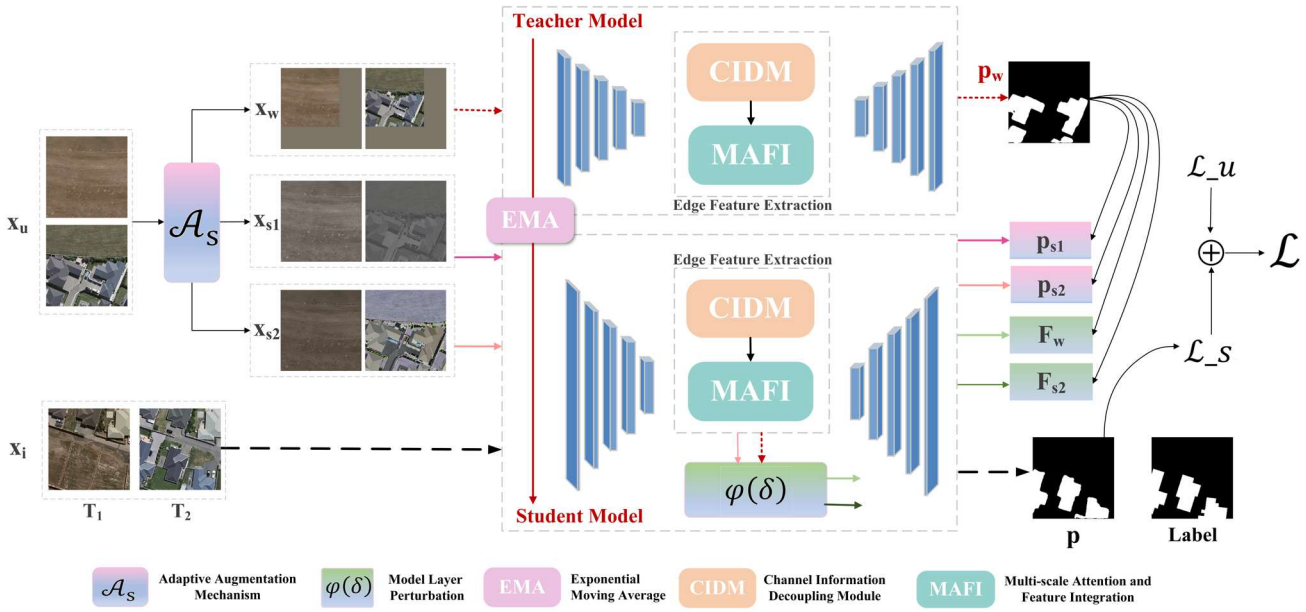


Fig. 1. Overview of the proposed SCAD-Net framework, which integrates a problem-specific architecture with a novel P2R-CL strategy. The framework employs a shared-weight DeepLabv3+ backbone for feature extraction from bitemporal images (T1 and T2). Two key modules are introduced: (1) the CIDM, which explicitly separates change information into structural and semantic streams to model boundaries and connectivity and (2) the MAFI module, which intelligently fuses these decoupled streams to suppress pseudochanges. The entire network is trained under the P2R-CL strategy, which computes losses at both the high-confidence pixel level and the refined region level to ensure learning stability.

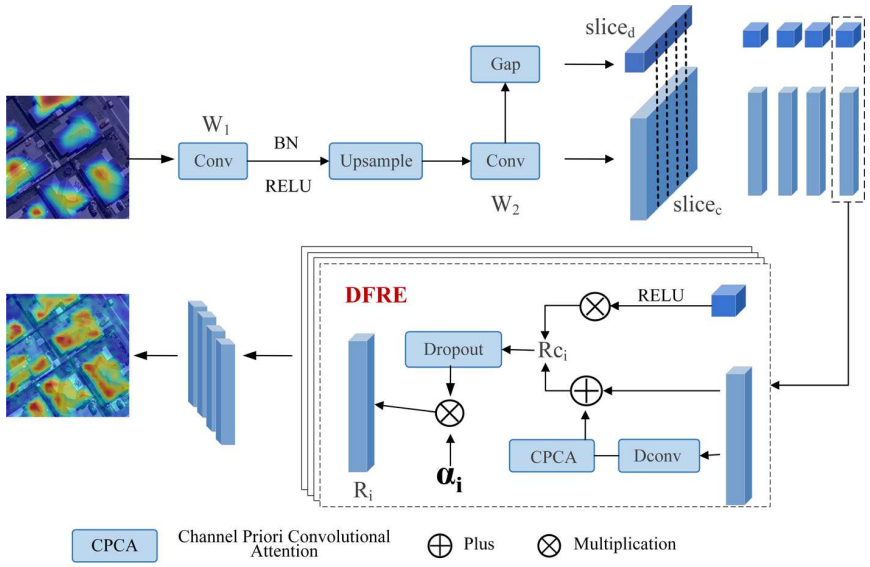


Fig. 2. Core principle is to first decouple semantic category features from directional structural features within a single processing stream. This separation allows the module to then specifically capture and enhance highly discriminative directional features. Finally, these refined directional cues are adaptively fused back with the semantic information to produce a feature representation that is robust to edge features and rich in structural detail.

pseudolabeling, and SemiCD, FPA-SSCD, C2F-SemiCD, and CutMix-CD are based on consistency learning. The methods were used to conduct experiments on the four datasets (WHU-CD, CDD, LEVIR-CD, and SH-CD) with four different training data ratios (5%, 10%, 20%, and 40%). The x% of the training set was used as labeled data, while the remaining (1−x%) of the training set was used as unlabeled data.

1) Comparison of the Quantitative Results: The results of each method are listed in Tables II–V. It can be clearly seen that the proposed method outperforms the state-of-the-art methods, achieving the best performance across all four datasets—CDD (see Table II), WHU-CD (see Table III), WHU-CD (see Table IV), and SHD (see Table V)—under all four different labeled data ratios, as shown in Fig. 6.

The comprehensive evaluation demonstrates that SCAD-Net establishes new performance benchmarks across four challenging datasets. The improvements are not only statistically significant but also technically interpretable and consistent. On the WHU-CD dataset, SCAD-Net consistently outperforms the competitive CutMix-CD baseline, achieving the

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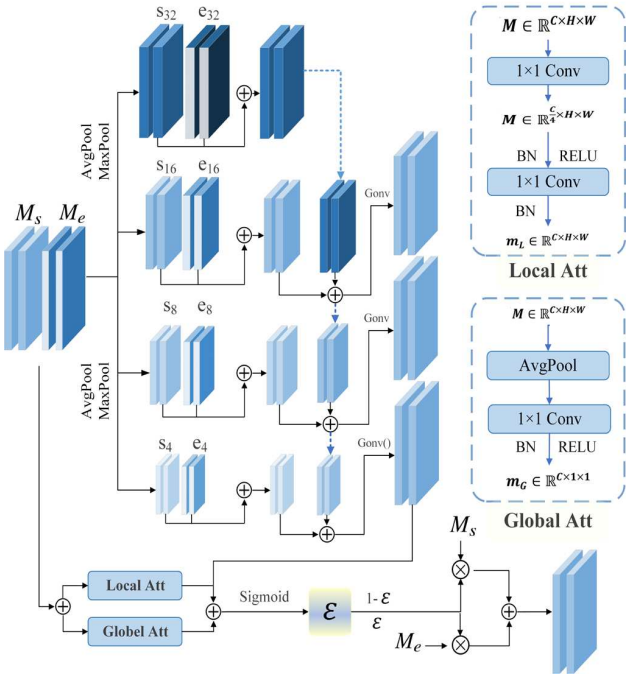


Fig. 3. Architecture of the proposed MAFI module. The module is designed to capture and fuse both semantic and texture information across multiple scales. It employs a dual-pooling strategy within four parallel context blocks, using average pooling for semantic contexts and max pooling for texture details.



Fig. 4. Overview of the SHD dataset. (a) T1 image. (b) T2 image. (c) Ground truth.

most substantial improvement of 1.96% in F1-score (90.79% versus 88.83%) under the 20% label setting. This improvement is particularly noteworthy given the already high baseline performance (F1 > 88%).

The technical efficacy of our approach is further validated through cross-dataset analysis. On LEVIR-CD, the 0.89% F1-score improvement (89.33% versus 88.44%) over C2F-SemiCD under the restrictive 5% labeling scenario demonstrates that our model effectively mitigates the feature ambiguity problem that typically degrades performance in low-supervision regimes. This is evidenced by comparing with competitive methods such as C2F-SemiCD, which experiences a 2.52% performance drop (90.96%–88.44%) when labeled data decrease from 40% to 5%, while SCAD-Net maintains better consistency with only a 1.10% decline (90.43%–89.33%), demonstrating the effectiveness of P2R-CL under limited supervision.

Further highlighting its robustness, SCAD-Net extends its competitive performance to the CDD dataset. By maintaining a leading position even in the presence of strong seasonal interference, the model proves its ability to generalize beyond urban scenes and effectively handle domain shifts caused by temporal variations. The SCAD-Net proves particularly effective on the SHD dataset, where the substantial 4.58% F1-score improvement (95.18% versus 90.60%) under a 40% labeling scenario over CutMix-CD indicates successful exploitation of both spatial and temporal consistency constraints to handle diverse change types.

Crucially, these performance gains are not isolated but demonstrate a consistent pattern of superiority across datasets with diverse characteristics—from urban changes (WHU-CD, LEVIR-CD, and SHD) to seasonal variations (CDD). This cross-dataset consistency demonstrates that our architectural innovations address fundamental, rather than dataset-specific limitations in semi-supervised CD, confirming the model's generalization capabilities.

2) Comparison of the Qualitative Results: Figs. 7–9 show the qualitative visualization results for the four datasets. To make the visualization results more intuitive, we include the TP (white), TN (black), FP (red), and FN (blue). From the visualization results, it can be seen that the proposed model significantly reduces the probability of false detection of false positive classes (red regions) and false negative classes (blue regions). This robustness is not only reflected in the quantitative metrics but also more intuitively reflected in the clear and accurate visualization results. It is particularly noteworthy that the proposed method performs well in dealing with changing regions of different spatial scales and complexity. Whether it is a cluster of densely arranged buildings (see Fig. 8) or sparsely distributed structures (see Fig. 9), SCAD-Net can accurately capture the change edges and achieve high-precision CD.

On all datasets, both AdvNet and S4GAN have serious problems with missed detection and false detection, and RCL and FPA-SSCD tend to fail in identifying small-scale changes, leading to higher missed detection rates for subtle variations [see Figs. 7(c) and 8(d)]. Although C2F-SemiCD and CutMix-CD achieve relatively acceptable CD results, they still have significant deficiencies in CD in the edge regions [see Figs. 7(a) and (b) and 4(a)]. In contrast, SCAD-Net exhibits an excellent performance, effectively recognizing both the edges and the subjects of the objects. These observations demonstrate the significant role of the CIDM in enhancing the perception of model edge information. C2F-SemiCD effectively detects change edges for small changes [see Fig. 8(b) and (c)], but, owing to the receptive field limitations, the detection of large objects is incomplete [see Fig. 4(a)], and the other methods tend to overlook the peripheral small target features [see Fig. 9(c) and (d)]. However, SCAD-Net overcomes these limitations and successfully detects these objects. This outstanding advantage can be attributed to the innovative design of the MAFI module, which significantly enhances the content integrity and improves the detection accuracy by effectively guiding the fusion of edge and semantic information.

C. Ablation Experiments

To validate the effectiveness of the proposed modules in SCAD-Net, we conducted ablation experiments for each module under different labeled data proportions on the LEVIR-CD and CDD datasets. P2R-CL is the basic semi-supervised

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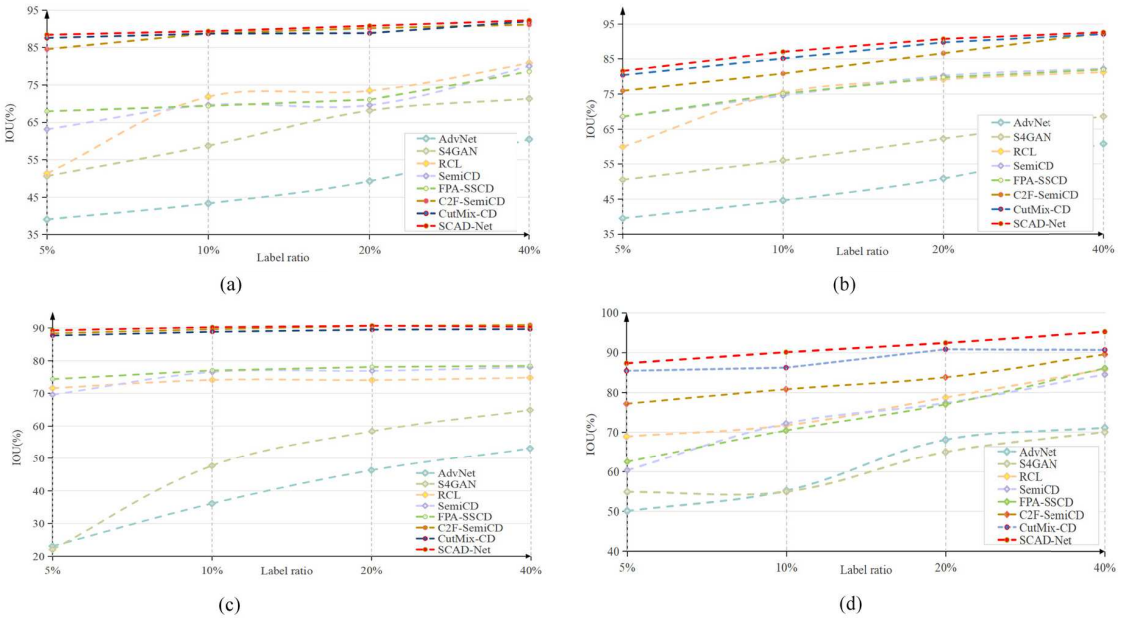


Fig. 5. Comparison between the state-of-the-art methods and the proposed SCAD-Net framework. (a) WHU-CD. (b) CDD. (c) LEVIR-CD. (d) SHD.

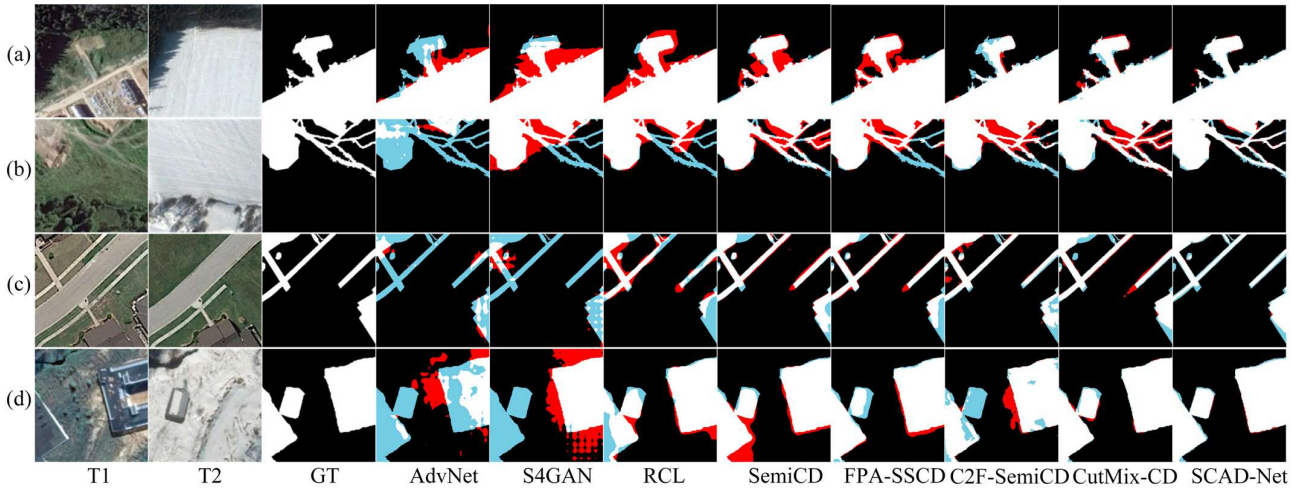


Fig. 6. Visualization results of the different methods for the CDD dataset with 5% labeled data. TP (white), TN (black), FP (red), and FN (blue). (a)–(d) Denote four different regions from the CDD dataset.

network that we have proposed. As shown in Table VI, both the CIDM and MAFI modules bring significant performance improvements. A trend emerges from the table: the performance margin gained by our modules widens as more labeled data are available, peaking at the 20% and 40% settings. This phenomenon indicates that our modules are not merely a compensation for data scarcity but rather powerful enhancers of feature representation. When the baseline model has access to a richer supervisory signal, it learns a more solid feature foundation. Our modules can then capitalize on these well-formed features to more effectively capture intricate contextual dependencies and integrate multilevel information.

We also conducted a visual analysis through the heatmaps output by the three stages of the network (see Figs. 10 and 11), which shows the heatmaps of the features in the three stages of the network from shallow to deep, including the semantic features, the edge features of the CIDM output, and the fusion features of the MAFI output. In comparison, it can be observed that, in the semantic feature stage, the network focuses on the overall region, but the edge information is more ambiguous. In the edge feature extraction stage, the CIDM module can extract the edge information effectively, which significantly enhances the detail performance for the edge regions. In the feature fusion stage, the MAFI module retains the overall structure while highlighting the edge information through the effective fusion of semantic and edge features, which improves the network's ability to recognize the real changed regions.

D. Discussion of the Neighborhood Design

To reduce the network complexity and improve the computational efficiency, we simplified the traditional eight-neighborhood connectivity design to a four-neighborhood

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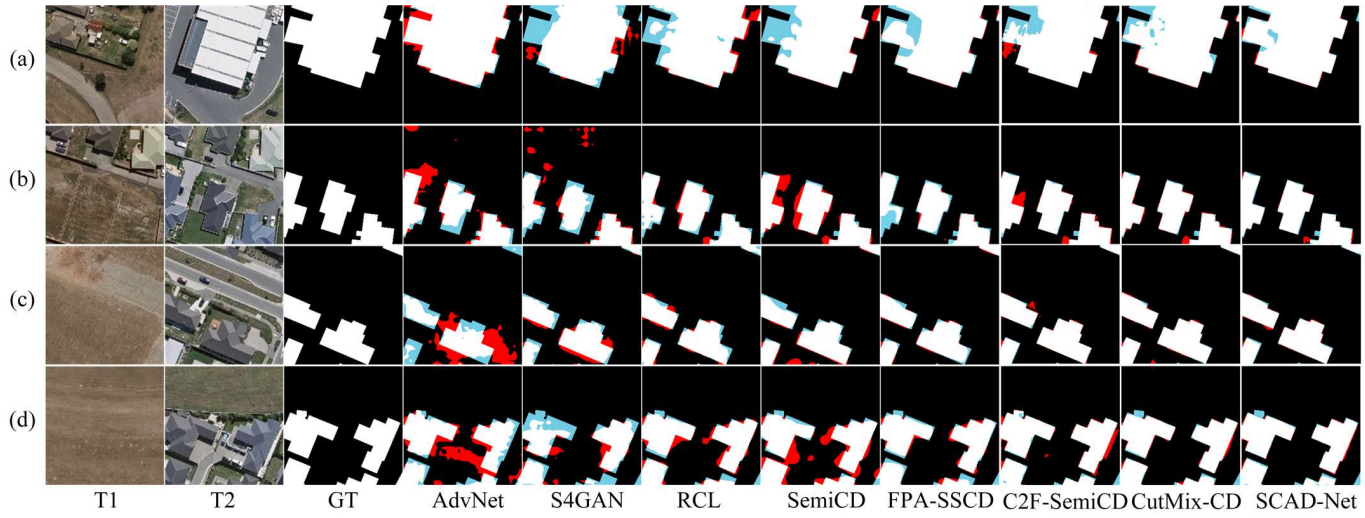


Fig. 7. Visualization results of the different methods for the WHU-CD dataset with 10% labeled data. TP (white), TN (black), FP (red), and FN (blue). (a)–(d) Denote four different regions from the WHU-CD dataset.

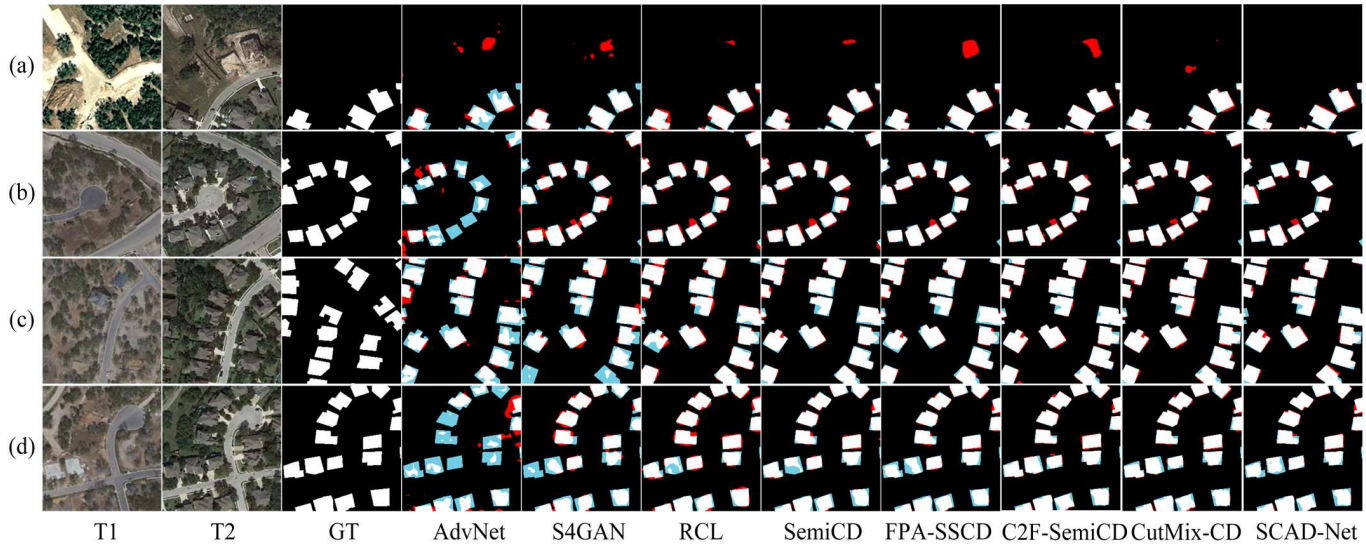


Fig. 8. Visualization results of the different methods for the LEVIR-CD dataset with 20% labeled data. TP (white), TN (black), FP (red), and FN (blue). (a)–(d) Denote four different regions from the LEVIR-CD dataset.

TABLE VI
ABLATION EXPERIMENTS ON THE LEVIR-CD AND CDD DATASETS

Dataset	Module			5%			10%			20%			40%		
	P2R-CL	CIDM	MAFI	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
LEVIR-CD	√			88.06	87.46	78.67	87.11	86.48	77.17	88.03	87.44	78.62	88.18	87.59	78.85
	√	√		88.18	87.58	78.86	89.99	89.47	81.79	88.56	87.97	79.47	90.23	89.73	82.20
	√	√	√	**89.33**	**88.78**	**80.72**	**90.31**	**89.81**	**82.34**	**90.76**	**90.27**	**83.08**	**90.43**	**89.94**	**82.53**
CDD	√			79.61	77.30	66.13	85.29	83.53	74.35	87.71	86.22	78.11	89.65	88.35	81.24
	√	√		80.61	78.43	67.52	86.42	84.77	76.08	89.12	87.77	80.38	91.22	90.11	83.86
	√	√	√	**81.60**	**79.50**	**68.92**	**87.05**	**85.47**	**77.07**	**90.75**	**89.59**	**83.07**	**92.63**	**91.68**	**86.27**

connectivity design. To validate the effectiveness of this design, we conducted experiments on the WHU-CD and SHD datasets to compare the accuracy of the eight- and four-neighborhood models, with the results summarized in Table VII. The findings indicate that, while the eight-neighborhood model achieves a slightly higher accuracy at

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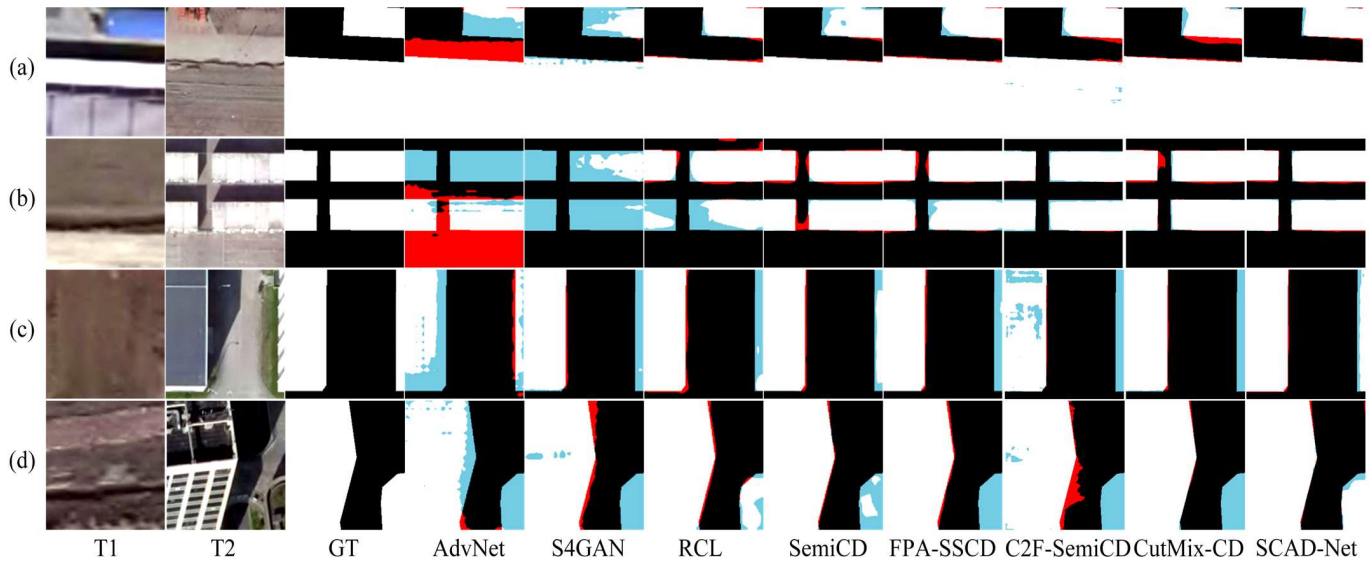


Fig. 9. Visualization results of the different methods for the SHD dataset with 40% labeled data. TP (white), TN (black), FP (red), and FN (blue). (a)–(d) Denote four different regions from the SHD dataset.

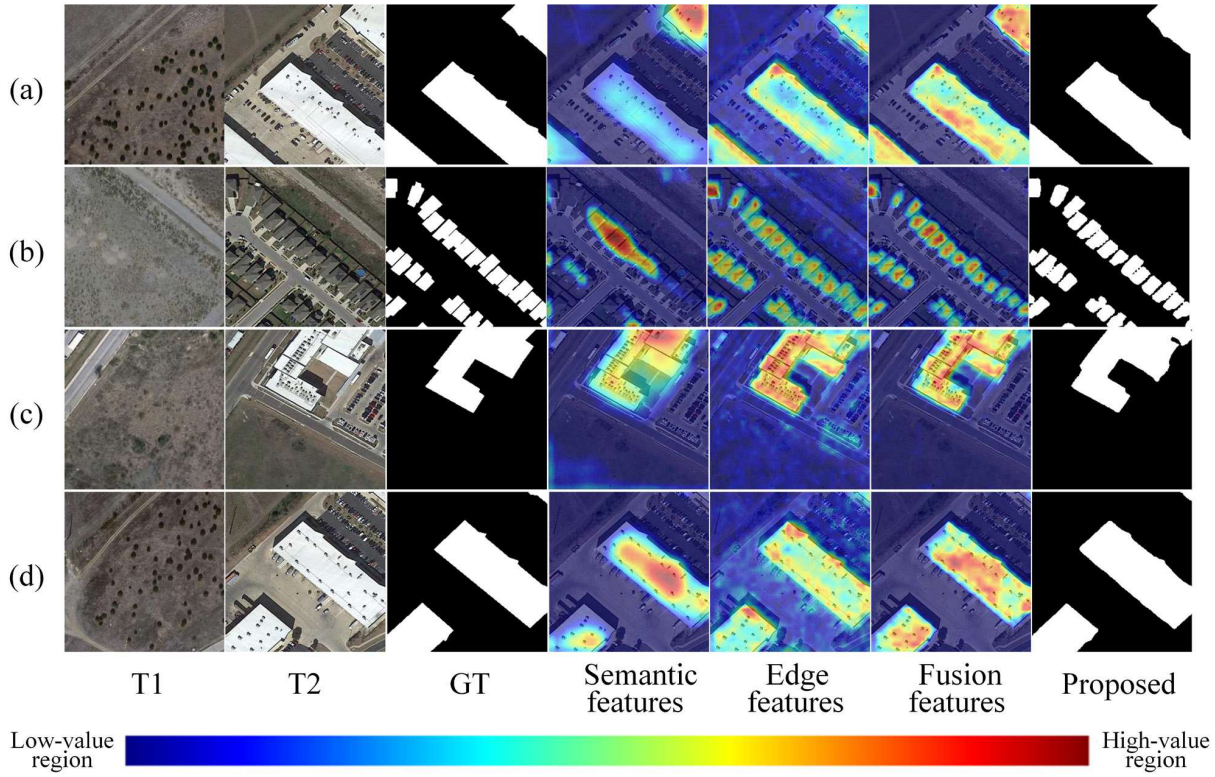


Fig. 10. Heatmap visualization of the feature representations across the three stages (semantic feature, edge feature extraction, and feature fusion) in the network using 40% labeled data from the LEVIR-CD dataset. (a)–(d) Denote four different regions from the LEVIR-CD dataset.

lower label-to-data ratios, it exhibits a significantly higher computational complexity. In contrast, the four-neighborhood model demonstrates comparable or superior accuracy in most cases while substantially reducing the number of parameters and computational overhead, thereby showcasing greater efficiency and practicality. These results confirm that the transition to a four-neighborhood design does not compromise the preservation of effective spatial structure information. On the contrary, it enhances the network's ability to efficiently capture edges and spatial structural features in changing regions by eliminating redundant information. Furthermore, the simplified four-neighborhood design reduces the network complexity and offers increased flexibility for future expansion and deployment.

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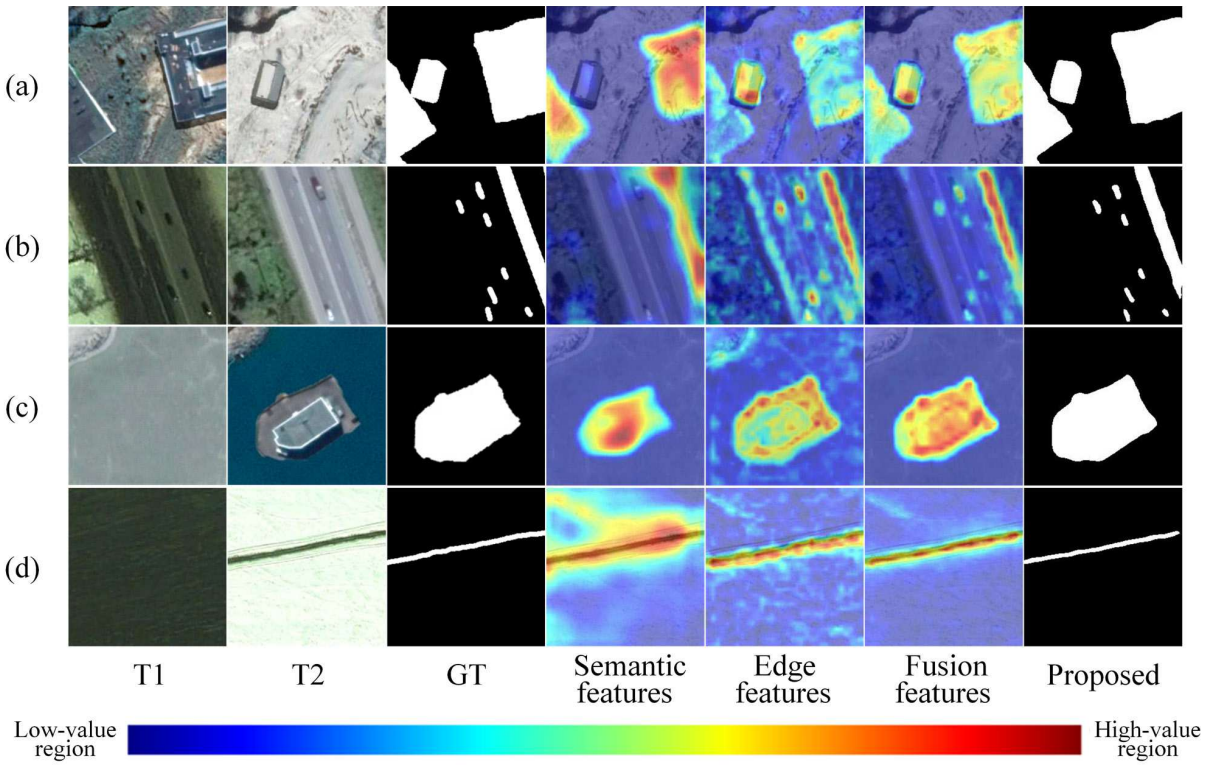


Fig. 11. Heatmap visualization of the feature representations across the three stages (semantic feature, edge feature extraction, and feature fusion) in the network using 40% labeled data from the CDD dataset. (a)–(d) denote four differnet regions from the CDD dataset.

TABLE VII

QUANTITATIVE PERFORMANCE OF THE CD MODELS USING THE FOUR-[CIDM(4)] AND EIGHT-NEIGHBORHOOD [CIDM(8)] CONNECTIVITY MODELS ON THE WHU-CD AND SHD DATASETS UNDER VARIOUS LABELED DATA RATIOS (5%, 10%, 20%, AND 40%)

Dataset	Model	FLOPs (G)	Params (M)	5%			10%			20%			40%		
				F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
WHU-CD	CIDM(8)	55.89	31.56	85.00	84.42	73.91	89.94	89.54	81.72	88.15	87.66	78.80	89.94	89.54	817.2
	CIDM(4)	55.67	31.53	**88.40**	**87.95**	**79.22**	**89.35**	**88.93**	**80.75**	**90.79**	**90.41**	**83.13**	**92.25**	**91.94**	**85.61**
SHD	CIDM(8)	55.89	31.56	82.43	81.03	70.11	88.69	87.76	79.68	91.14	90.42	83.73	93.68	93.15	88.10
	CIDM(4)	55.67	31.53	**87.32**	**86.26**	**77.49**	**90.10**	**89.29**	**81.98**	**92.42**	**91.78**	**85.90**	**95.18**	**94.77**	**90.79**

E. Discussion on the SSL Strategy

P2R-CL is founded upon the principle of consistency-based regularization, which introduces a critical departure to address the limited labeled sample challenges of remote sensing CD. Specifically, we analyzed a key limitation factor in confidence thresholding, as used in frameworks such as UniMatch, which fails to adapt to the dynamic evolution of model confidence in complex, imbalanced remote sensing environments. A static threshold is either too permissive in early training stages, admitting significant label noise, or too restrictive in later stages, underutilizing valuable unlabeled data.

The main innovation of P2R-CL is the replacement of this static mechanism with a dynamic, curriculum-guided learning strategy. This curriculum is realized through our concrete " pixels to regions" consistency framework. Initially, the model learns from high-confidence pixels, establishing a robust foundation free from significant label noise. As the model matures, the learning objective transforms to enforcing consistency at the level of semantically coherent regions. This crucial shift enables the model to refine imprecise pseudolabel boundaries and learn high-level concepts, rather than memorizing noise. This core curriculum is further enhanced by multilevel perturbations, which compel the model to learn these invariant semantics by augmenting it at both the input-image and internal-feature levels. This challenges the model to learn invariant semantics. Table VIII demonstrates that synergy is the key to the performance gains.

F. Cost Analysis of Time, Storage, and Computation

In this section, we analyze the cost of the model in detail, including factors such as storage, computational complexity, and inference time. Specifically, we used a pair of images of size $3 \times 256 \times 256$ to calculate the FLOPs, parameters, and inference times. All the images from the public datasets used in the previous experiments were consistently of size $3 \times 256 \times 256$. FLOPs refers to the computational complexity of the model, in GFLOPs; Params refers to the size of the model parameter file, in Mb; and inference time refers to the time

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TABLE VIII

ABLATION STUDY ON THE SSL STRATEGY IN SCAD-NET. WE COMPARE OUR PROPOSED P2R-CL AGAINST THE UNIMATCH STRATEGY ON THE WHU-CD AND LEVIR-CD DATASETS UNDER VARIOUS LABELED DATA RATIOS (5%, 10%, 20%, AND 40%)

Dataset	Module	5%			10%			20%			40%		
		F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
WHU-CD	SCAD-Net(Unimatch)	84.17	83.51	72.67	87.97	87.47	78.53	84.69	84.12	73.44	89.50	89.07	81.00
	SCAD-Net(P2R-CL)	**88.40**	**87.95**	**79.22**	**89.35**	**88.93**	**80.75**	**90.79**	**90.41**	**83.13**	**92.25**	**91.94**	**85.61**
LEVIR-CD	SCAD-Net(Unimatch)	87.56	86.92	77.87	87.72	87.10	78.12	88.15	87.55	78.81	87.60	86.98	77.93
	SCAD-Net(P2R-CL)	**89.33**	**88.78**	**80.72**	**90.31**	**89.81**	**82.34**	**90.76**	**90.27**	**83.08**	**90.43**	**89.94**	**82.53**

TABLE IX

COMPARISON OF THE FLOPS, PARAMETERS, AND INFERENCE TIMES FOR THE DIFFERENT MODELS

	FLOPs (G)	Params (Mb)	Inference time (ms)
AdvNet [66]	**49.11**	45.34	**7.99**
S4GAN [67]	49.21	45.34	12.76
RCL [50]	73.23	46.85	21.80
SemiCD [68]	73.23	46.85	46.69
FPA-SSCD [51]	73.23	46.85	25.50
C2F-SemiCD [56]	62.10	**16.17**	23.54
CutMix-CD [69]	73.23	46.85	8.96
SCAD-Net	55.67	31.53	20.61

required for the model to process a pair of dual-phase images, in ms. The cost comparison results are listed in Table IX.

It is important to note that RCL, SemiCD, FPA-SSCD, and CutMix-CD share the same model architecture, despite employing different semi-supervised strategies, and, therefore, have identical parameter counts and computational requirements. In contrast, AdvNet has the smallest number of parameters and the fastest inference speed, while the computational efficiency and inference time of SCAD-Net are lower than those of CutMix-CD, RCL, SemiCD, and FPA-SSCD. Nevertheless, when considering the model accuracy comprehensively, SCAD-Net demonstrates a significant improvement in accuracy over AdvNet and S4GAN. In summary, the proposed SCAD-Net framework achieves a notable efficiency in both training and inference while offering substantial improvements in accuracy, making it an effective and reliable model.

G. Limitations and Future Work

Our work establishes a robust and effective framework for SSCD. While SCAD-Net outperforms all listed prior methods, its current scope is designed for single-modality, bitemporal imagery. This successful foundation provides a clear path for future research into more complex and diverse scenarios.

Furthermore, we plan to extend this research in three key directions. First, we will adapt the framework for multisource data by engineering more sophisticated fusion mechanisms. This involves not only spatiotemporal alignment [73] but also advanced multimodal interaction networks [74], [75] designed to bridge the semantic gap between disparate data types. Second, to move beyond local feature analysis, we will investigate several advanced architectures capable of modeling global, object-level relationships. These include graph neural networks (GNNs) [76], vision transformers (ViT) [28], [75], and emerging state-space models such as Mamba [24], [30], all of which are effective at capturing long-range spatial dependencies. Finally, to further enhance data efficiency in scenarios of extreme label scarcity, we will integrate a richer suite of data augmentation techniques [77], [78], [79].

V. CONCLUSION

In this article, we have proposed SCAD-Net to overcome two critical bottlenecks in SSCD: the cyclical propagation of errors from noisy pseudolabels and the loss of fine-grained boundary details. The proposed approach introduces three key innovations: 1) the CIDM module, which precisely constructs the spatial relationships to enhance the edge detection accuracy; 2) the MAFI module, which effectively fuses fine-grained edge details with high-level semantic context across multiple scales; and 3) P2R-CL, which introduces a pixel-to-region contrastive learning strategy to refine noisy pseudolabels and guide the model toward learning more discriminative features. Extensive experiments on four publicly available datasets demonstrated that SCAD-Net consistently outperforms the state-of-the-art methods across various labeled data ratios, achieving remarkable improvements in both the quantitative metrics and qualitative visual results. These comprehensive evaluations validated the effectiveness of the architectural design and established SCAD-Net as a robust solution for semi-supervised CD tasks. Future work will aim to extend this framework by incorporating multisource data and more advanced feature representation techniques, as discussed in Section IV-G.

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